Esequiel Linares

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Education

Purdue University West Lafayette, IN
Bachelor of Science in Computer Science *Expected Graduation, May 2027*

Experience

Undergraduate Researcher – Multimodal Generative AI September 2025 – Present
Purdue University *West Lafayette, IN*

- Led research on Duet Dance dataset (10.3 hrs, 15 genres, 30 dancers, 10K+ annotations, 4K sequences), achieving up to 6× accuracy gains in duet generation.
- Built PyTorch pipelines for Text-to-Duet and Dance Accompaniment, improving models by 60–500%.
- Analyzed VR/XR and multimodal embeddings, identifying 5 use cases and contributing to a research paper.

Capital One First Gen Focus May 2025 – July 2025
Participant (Top 10% Selected) *Chicago, IL (Virtual)*

- Chosen from 1,000+ applicants for an 8-week program focused on empowering first-generation students.
- Executed 10+ hours of coding workshops on LeetCode prep and software engineering career navigation.
- Advanced backend and ML skills through Capital One SWE mentorship, building 3 projects including a multilayer perceptron that improved name-generation accuracy by 25% and diversity by 40%.

Programming Instructor August 2022 – June 2025
Self-Employed *North Riverside, IL (Remote)*

- Taught 15 students in Java/Python, raising averages to 90% in computer science courses.
- Designed and delivered engaging lessons, resulting in improved academic performance and higher school scores.
- Mentored projects, driving a 20% increase in exam performance.

Projects

Reachy Mini (HCI) | *C++, Hugging Face, PyTorch, WebRTC, RunPod (Cloud GPUs), CUDA* August 2025 – Present

- Architected a humanoid-inspired robotic assistant that fuses speech, vision, and decision-making through a low-latency cloud–edge pipeline combining Raspberry Pi 5 with GPU servers.
- Integrated Hugging Face Transformers for multimodal emotion and intent recognition, elevating interaction accuracy by 40% and enabling adaptive, context-aware dialogue.
- Engineered WebRTC streaming and optimized C++ drivers for Pi 5 speaker/camera modules, reducing I/O delays by 25% and sustaining sub-100ms end-to-end response times.

Autograd Engine (Deep Learning) | *Python, NumPy, PyTorch, Matplotlib, Google Colab* May 2025 – June 2025

- Built a reverse-mode autodiff engine from scratch, mimicking PyTorch's autograd.
- Trained a 2-layer multi-layer perceptron to 100% accuracy on synthetic data.
- Engineered test coverage to ensure accurate gradient flow and output predictions.

AirCursor (Computer Vision) | *Google Mediapipe, OpenCV, NumPy, Google Colab* August 2024 – November 2024

- Built a CNN on 1,440 video frames, reaching 95% accuracy in hand-gesture recognition.
- Accelerated data processing, cutting training time to 5 minutes and enabling rapid gesture creation.
- Implemented a low-latency cursor control system with PyAutoGUI and OpenCV, enhancing real-time interaction.

Shop22 (E-commerce) | *TypeScript, Stripe API, Sass, Payload CMS, Next 14, Docker* June 2024 – December 2024

- Developed an e-commerce platform with React and Payload CMS, enhancing shopping with customized product listings.
- Implemented secure payment processing with Stripe API, optimizing transaction flow and customer trust.
- Configured Docker for seamless deployment, ensuring scalable and maintainable application architecture.

Technical Skills

Languages: Python, Java, C++, C, C#, JavaScript, TypeScript, SQL, HTML, Bash
Web & Styling Tools: CSS, Sass, Tailwind CSS
Frameworks & Libraries: React, Node.js, Next.js, Spring Boot, Java Swing, JavaFX, React Router, PyTorch, Hugging Face, NumPy, Pandas, Matplotlib, Mediapipe, PyAutoGUI, Apache Commons, CUDA, JUnit, pytest
Developer Tools: Git, Docker, AWS, VS Code, Cursor AI, Visual Studio, CLion, IntelliJ, PyCharm, Google Cloud Platform, Linux/Unix, PostgreSQL, RunPod, Google Colab Pro